

STUDENT VENTURE CAPITAL GRANT (SVCG) 2026

MechMind AI

Smart Heavy Construction Machinery Diagnostics & Predictive Telematics

Bridging the gap between IoT condition monitoring and mechanics via WhatsApp

75% Diagnostic Accuracy

56-Question Blind Benchmark

Validated across hydraulic, electrical & engine faults

NGN 35,000 Unit Hardware Cost

ESP32 + Tri-Axial Sensors

Replaces NGN 8M+ imported proprietary diagnostic tools

90-Second WhatsApp Response

Real-Time Telematics & RAG

Delivers verified OEM repair procedures & part numbers

NGN 500,000 — NGN 2,000,000 Lost Daily Per Stalled Machine

Equipment downtime in Nigeria's NGN 6.4T infrastructure sector creates severe project delays, idle labour, and wasted repair costs.

1. The Language & Tech Barrier

Over 60% of excavators and loaders entering Nigeria are Chinese brands (XCMG, SANY, Shantui). Error codes and technical workshop manuals are in Chinese or dense engineering notation that local mechanics cannot decipher.

2. Trial-and-Error Misdiagnosis

With no affordable diagnostics available, technicians guess faults by trial and error, replacing expensive functional parts blindly. This causes recurring breakdowns and safety hazards on road and building projects.

3. High Cost of Foreign Scanners

OEM diagnostic tools (Caterpillar Product Link, Komatsu KOMTRAX) cost NGN 5M–NGN 8M+, require specialized subscriptions, and only work on a single brand. Small/medium contractors are left completely excluded.

Layer 1: The MechMind Node (IoT Hardware)

- Magnetic Clamping:** Installs in 30 seconds on any machine surface without drilling.
- Multi-Modal Sensing:** Tri-axial vibration (ADXL345), surface temperature (DS18B20), acoustics (INMP441).
- Edge Intelligence:** ESP32 calculates real-time FFT vibration spectrum and flags mechanical anomalies.
- Ultra-Low Cost:** Bills at under \$15 (NGN 25,000) in components vs NGN 8M imported units.
- Rechargeable Power:** Integrated 18650 Li-ion battery + solar/alternator charging circuit.

Layer 2: WhatsApp AI Diagnostic Assistant

- Zero Friction Access:** No apps to download. Mechanics interact entirely through WhatsApp (+234 701 029 9562).
- Multi-Modal Inputs:** Accepts typed fault codes, voice notes (Whisper speech-to-text), and broken component photos.
- RAG Grounded Intelligence:** Responses verified against indexed OEM manuals (XCMG, SANY, Caterpillar, Komatsu).
- Actionable Output:** Delivers probable root cause, step-by-step repair guide, and OEM part numbers in 90s.
- Fleet Cloud Dashboard:** Web console (mechmind-ng.vercel.app) gives fleet managers real-time fleet health.

75.0%

Diagnostic Accuracy

Evaluated on 56-question blind test suite across hydraulics, electrical & power

< 90 Sec

Average Triage Speed

Cuts diagnostic downtime from 2+ days waiting for specialists down to minutes

Live & Public

WhatsApp Assistant

Active WhatsApp bot (+234 701 029 9562) handling multi-turn diagnostics

Deployed Cloud

Fleet Telemetry Hub

Production dashboard live at mechmind-ng.vercel.app with interactive 3D model

Real-World Field Validation & Technical Readiness

- 1. **Physical Hardware Assembled & Tested:** Sensor nodes assembled with ESP32, ADXL345 accelerometer, and DS18B20 temperature probes, encased in 3D-printed magnetic shells.
- 2. **Nigerian Society of Engineers (NSE) Alignment:** Invited to the 15th China Innovation & Entrepreneurship Competition (AI + Construction Machinery) supported by XCMG West Africa.
- 3. **Full Multi-Modal AI Pipeline Deployed:** FastAPI backend, pgvector RAG manual indexing, Twilio WhatsApp webhook, and real-time WebSocket dashboard.
- 4. **Commercial Entity Registered:** Pullup Events (CAC Registration BN-9846744) covering Technology & Software services.

NGN 6.4 Trillion

Nigerian Construction Sector

Rapid infrastructure expansion under the Renewed Hope Agenda

80,000+

Active Heavy Machines

Excavators, bulldozers, wheel loaders, and dump trucks operating nationwide

15,000+

Contractors & Rental Hubs

Civil contractors, plant hire operators, and quarry sites needing diagnostics

Target Customer Segments & Deployment Strategy

Civil Engineering Contractors

Mid-sized contractors with 5–25 machines who cannot afford full-time diagnostic specialists. MechMind prevents costly job-site penalties.

Heavy Equipment Plant Hire Firms

Rental operators who lose revenue when machines break down on client sites. Real-time telemetry protects rental assets from operator abuse.

Independent Machinery Technicians

Over 20,000 on-ground mechanics across Nigeria using WhatsApp daily to triage complex electronic and hydraulic issues.

1. Hardware Node Sales

NGN 35,000 – NGN 50,000

One-Time Purchase

Rugged ESP32 sensor node sold with healthy 40% gross margin over \$15 BOM. Magnetic clip-on design makes self-installation effortless for fleet operators.

2. SaaS Telemetry Subscription

NGN 15,000 / machine

Monthly Recurring (MRR)

Continuous live telemetry monitoring, automated anomaly alerts, fleet health analytics dashboard, and unlimited priority WhatsApp AI diagnostic queries.

3. Spare Parts Commission

2% – 3% per order

Marketplace Transaction Fee

Direct matchmaking with certified OEM parts suppliers in Lagos and Abuja. Mechanics order verified replacement parts directly via the WhatsApp diagnostic flow.

Mubarak Raji Babatunde

Project Leader & AI / Software Engineer

LASU Civil Engineering (100L | Matric: 230241036)

Full-stack and AI developer. Built the MechMind RAG pipeline, WhatsApp bot, telemetry API, and Next.js cloud dashboard. Won 1st Place at Vibranium Valley x PowerTech Grand Challenge.

Olawale Oladele

Hardware Design & Machine Learning Engineer

LASU Mechanical Engineering (100L | Matric: 230221043)

CSWP SolidWorks Professional certified with 4 CSWPA specializations and Ansys FEA credentials. Designed the 3D-printed enclosure, sensor circuit schematics, and firmware algorithms.

Oni Mujeeb Ramadan

Product Manager & Operations Lead

LASU Civil Engineering (100L | Matric: 230241032)

NUESA Engineering Scholar of the Year (2026), Man of Excellence Award (LASUSU). Manages site field tests, equipment operator interviews, and commercial partner outreach across Lagos.

Months 1 – 3

Pilot Hardware Production
& Lab Benchmark

NGN 12,500,000 (25%)

- Manufacture first 50 production-grade SMT PCB sensor nodes.
- Expand manual indexing to 20 Caterpillar & XCMG models.
- Deploy pilot units across 10 machines in Lagos construction sites.
- Achieve 80%+ diagnostic accuracy on expanded test set.

Months 4 – 6

Fleet Deployment & WhatsApp
Scaling

NGN 15,000,000 (30%)

- Onboard 25 commercial civil engineering contractors and plant hire fleets.
- Implement automated GSM/cellular telemetry fallback for off-grid sites.
- Integrate direct spare part ordering through WhatsApp.
- Target NGN 1.5M monthly recurring revenue (MRR).

Months 7 – 9

Multi-Brand Expansion & Model
Tuning

NGN 12,500,000 (25%)

- Index 35+ equipment brands (Komatsu, Shantui, SANY, Hitachi).
- Train localized multimodal vision model for heavy machinery wear detection.
- Expand field operations into Ogun State and Abuja infrastructure corridors.
- Secure distribution partnerships with 3 major equipment dealerships.

Months 10 – 12

National Commercial Scale

NGN 10,000,000 (20%)

- Scale to 250 active hardware sensor nodes across 50 fleets.
- Launch predictive remaining useful life (RUL) enterprise dashboard.
- Reach financial sustainability with NGN 5M+ monthly recurring revenue.
- Position MechMind AI as Nigeria's standard equipment telemetry system.

Empowering Nigeria's Infrastructure with Intelligent AI

MechMind AI — Student-Led STEMM Innovation Ready for Commercial Scale

Verify MechMind AI Live Today

- **WhatsApp Diagnostic Bot:** [+234 701 029 9562](tel:+2347010299562)
- **Live Telemetry Dashboard:** <https://mechmind-ng.vercel.app>
- **Open-Source Codebase:** github.com/DevMubarak1/mechmind-ai
- **CAC Business Registration:** [Pullup Events \(CAC BN-9846744\)](#)